SparseRandomProjections_v2

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1 Assignment #1

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1.1 Datasets

Dataset 1: Kaggle Pokémon Dataset https://www.kaggle.com/datasets/rounakbanik/pokemon/

Dataset 2: Fashion-MNIST (Zalando Research) https://www.kaggle.com/datasets/zalando-research/fashionmnist

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1.2 Introduction

We use two datasets to study how sparse random projections behave across fundamentally different types of data: a low-dimensional tabular dataset with engineered features (Pokémon) and a higher-dimensional image dataset with raw pixel features (Fashion-MNIST).

The **Pokémon dataset** consists of engineered attributes such as type-matchup multipliers, base stats, and categorical flags. These features do not live in a natural geometric space — Euclidean distance between two Pokémon vectors has no obvious physical meaning, and many features are correlated or partially redundant (e.g. type interactions with one-hot encoded types). The task is binary classification: predict whether a Pokémon is legendary. Legendary Pokémon are rare, powerful entities within the Pokémon universe, typically unique, non-evolving, and tied to specific in-game events or lore — which also makes the dataset strongly class-imbalanced (~9% legendary).

The **Fashion-MNIST dataset** consists of 28×28 grayscale images of clothing items across 10 classes, represented as 784-dimensional pixel-intensity vectors. Unlike the Pokémon features, these vectors live in a genuine Euclidean space: pairwise distances reflect visual similarity, and the data is known to concentrate on a lower-dimensional manifold within the ambient 784-dimensional space. This makes it a natural setting for random projections, which the Johnson–Lindenstrauss lemma guarantees will approximately preserve pairwise distances.

[21]:
```
# import general libraries
import pandas as pd
import kagglehub
import os
import numpy as np
```

```
import matplotlib.pyplot as plt

# import sklearn libraries
from sklearn import metrics
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.svm import LinearSVC
from sklearn.random_projection import SparseRandomProjection,␣
↪johnson_lindenstrauss_min_dim
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score, precision_score, recall_score,␣
↪f1_score

%matplotlib inline

import warnings
warnings.filterwarnings('ignore')
```

[22]:

```
# Shared helper functions for random projection experiments

def make_dims(max_dim, n_points=20):
    """Log-spaced dimensions: denser at low k, sparser at high k."""
    return np.unique(np.geomspace(2, max_dim, n_points).astype(int))


def run_rp_experiment(X_train, X_test, y_train, y_test, dims,
                      realizations=15, max_iter=2000, f1_avg='binary'):
    """Run sparse random projection + LinearSVC for each dimension,
    repeated over multiple realizations."""
    acc_all = np.zeros((realizations, len(dims)))
    f1_all  = np.zeros((realizations, len(dims)))

    for j, dim in enumerate(dims):
        for i in range(realizations):
            rp = SparseRandomProjection(n_components=dim, random_state=i)
            X_train_rp = rp.fit_transform(X_train)
            X_test_rp  = rp.transform(X_test)

            model = LinearSVC(max_iter=max_iter, random_state=42)
            model.fit(X_train_rp, y_train)
            y_pred = model.predict(X_test_rp)

            acc_all[i, j] = accuracy_score(y_test, y_pred)
            f1_all[i, j]  = f1_score(y_test, y_pred, average=f1_avg,␣
↪zero_division=0)
```

```
        print(f"  k={dim:4d}:  acc = {acc_all[:, j].mean():.4f} +/- {acc_all[:,␣
↪j].std():.4f}  |  "
              f"F1 = {f1_all[:, j].mean():.4f} +/- {f1_all[:, j].std():.4f}")

    return {
        'mean_acc': acc_all.mean(axis=0), 'std_acc': acc_all.std(axis=0),
        'mean_f1':  f1_all.mean(axis=0),  'std_f1':  f1_all.std(axis=0),
        'single_acc': acc_all[0],  # first realization for single-vs-ensemble␣
↪plot
    }


def plot_rp_results(dims, results, baseline_acc, baseline_f1, title,␣
↪realizations):
    """Standard 4-panel plot: accuracy, F1, variability, single vs. ensemble."""
    fig, axes = plt.subplots(2, 2, figsize=(12, 10))

    # Accuracy
    axes[0, 0].plot(dims, results['mean_acc'], 'b-o', markersize=4, label='RP␣
↪mean accuracy')
    axes[0, 0].fill_between(dims, results['mean_acc'] - results['std_acc'],
                            results['mean_acc'] + results['std_acc'], alpha=0.2)
    axes[0, 0].axhline(baseline_acc, color='r', linestyle='--',␣
↪label=f'Baseline: {baseline_acc:.3f}')
    axes[0, 0].set(xlabel='Projected dimensions k', ylabel='Accuracy',
                   title=f'{title}: Accuracy vs. Dimension')
    axes[0, 0].set_ylim(0, 1)
    axes[0, 0].legend()

    # F1-score
    axes[0, 1].plot(dims, results['mean_f1'], 'g-o', markersize=4, label='RP␣
↪mean F1')
    axes[0, 1].fill_between(dims, results['mean_f1'] - results['std_f1'],
                            results['mean_f1'] + results['std_f1'], alpha=0.2,␣
↪color='g')
    axes[0, 1].axhline(baseline_f1, color='r', linestyle='--', label=f'Baseline:
↪ {baseline_f1:.3f}')
    axes[0, 1].set(xlabel='Projected dimensions k', ylabel='F1-score',
                   title=f'{title}: F1-score vs. Dimension')
    axes[0, 1].set_ylim(0, 1)
    axes[0, 1].legend()

    # Variability
    axes[1, 0].plot(dims, results['std_acc'], 'b-o', markersize=4,␣
↪label='Accuracy std')
```

```
    axes[1, 0].plot(dims, results['std_f1'], 'g-o', markersize=4, label='F1␣
↪std')
    axes[1, 0].set(xlabel='Projected dimensions k', ylabel='Standard deviation',
                   title=f'{title}: Metric Variability across Projections')
    axes[1, 0].legend()

    # Single vs. Ensemble
    axes[1, 1].plot(dims, results['single_acc'], 'c-', alpha=0.7, linewidth=1,␣
↪label='Single realization')
    axes[1, 1].plot(dims, results['mean_acc'], 'b-', linewidth=2.5,␣
↪label=f'Averaged ({realizations} runs)')
    axes[1, 1].fill_between(dims, results['mean_acc'] - results['std_acc'],
                            results['mean_acc'] + results['std_acc'], alpha=0.
↪15, color='b')
    axes[1, 1].axhline(baseline_acc, color='r', linestyle='--',␣
↪label=f'Baseline: {baseline_acc:.3f}')
    axes[1, 1].set(xlabel='Projected dimensions k', ylabel='Accuracy',
                   title=f'{title}: Single Run vs. Ensemble Average')
    axes[1, 1].set_ylim(0, 1)
    axes[1, 1].legend()

    plt.tight_layout()
    plt.show()
```

1.3 1. Pokémon Dataset

[23]:
```
# load pokemon dataset
path = kagglehub.dataset_download("rounakbanik/pokemon")
df = pd.read_csv(os.path.join(path, "pokemon.csv"))
```

[24]:
```
# Exploratory Data Analysis (EDA)

# First look
pd.set_option('display.max_columns', None)
display(df.head())
pd.reset_option('display.max_columns')

# Check size
display(df.shape)

# Columns
display(df.columns)

# Data types
display(df.info())

# Missing values in %
```

```python
display((df.isna().sum() / len(df)) * 100)
```

```
                    abilities  against_bug  against_dark  against_dragon  \
0  ['Overgrow', 'Chlorophyll']          1.0           1.0             1.0
1  ['Overgrow', 'Chlorophyll']          1.0           1.0             1.0
2  ['Overgrow', 'Chlorophyll']          1.0           1.0             1.0
3     ['Blaze', 'Solar Power']          0.5           1.0             1.0
4     ['Blaze', 'Solar Power']          0.5           1.0             1.0

   against_electric  against_fairy  against_fight  against_fire  \
0               0.5            0.5            0.5           2.0
1               0.5            0.5            0.5           2.0
2               0.5            0.5            0.5           2.0
3               1.0            0.5            1.0           0.5
4               1.0            0.5            1.0           0.5

   against_flying  against_ghost  against_grass  against_ground  against_ice  \
0             2.0            1.0           0.25             1.0          2.0
1             2.0            1.0           0.25             1.0          2.0
2             2.0            1.0           0.25             1.0          2.0
3             1.0            1.0           0.50             2.0          0.5
4             1.0            1.0           0.50             2.0          0.5

   against_normal  against_poison  against_psychic  against_rock  \
0             1.0             1.0              2.0           1.0
1             1.0             1.0              2.0           1.0
2             1.0             1.0              2.0           1.0
3             1.0             1.0              1.0           2.0
4             1.0             1.0              1.0           2.0

   against_steel  against_water  attack  base_egg_steps  base_happiness  \
0            1.0            0.5      49            5120              70
1            1.0            0.5      62            5120              70
2            1.0            0.5     100            5120              70
3            0.5            2.0      52            5120              70
4            0.5            2.0      64            5120              70

   base_total capture_rate   classfication  defense  experience_growth  \
0         318           45    Seed Pokémon       49            1059860
1         405           45    Seed Pokémon       63            1059860
2         625           45    Seed Pokémon      123            1059860
3         309           45  Lizard Pokémon       43            1059860
4         405           45   Flame Pokémon       58            1059860

   height_m  hp     japanese_name        name  percentage_male  \
0       0.7  45  Fushigidane      Bulbasaur             88.1
1       1.0  60   Fushigisou        Ivysaur             88.1
2       2.0  80  Fushigibana       Venusaur             88.1
```

```
3       0.6  39      Hitokage    Charmander             88.1
4       1.1  58       Lizardo    Charmeleon             88.1

   pokedex_number  sp_attack  sp_defense  speed  type1   type2  weight_kg  \
0               1         65          65     45  grass  poison        6.9
1               2         80          80     60  grass  poison       13.0
2               3        122         120     80  grass  poison      100.0
3               4         60          50     65   fire     NaN        8.5
4               5         80          65     80   fire     NaN       19.0

   generation  is_legendary
0           1             0
1           1             0
2           1             0
3           1             0
4           1             0
```

```
(801, 41)
```

```
Index(['abilities', 'against_bug', 'against_dark', 'against_dragon',
       'against_electric', 'against_fairy', 'against_fight', 'against_fire',
       'against_flying', 'against_ghost', 'against_grass', 'against_ground',
       'against_ice', 'against_normal', 'against_poison', 'against_psychic',
       'against_rock', 'against_steel', 'against_water', 'attack',
       'base_egg_steps', 'base_happiness', 'base_total', 'capture_rate',
       'classfication', 'defense', 'experience_growth', 'height_m', 'hp',
       'japanese_name', 'name', 'percentage_male', 'pokedex_number',
       'sp_attack', 'sp_defense', 'speed', 'type1', 'type2', 'weight_kg',
       'generation', 'is_legendary'],
      dtype='str')
```

```
<class 'pandas.DataFrame'>
RangeIndex: 801 entries, 0 to 800
Data columns (total 41 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   abilities          801 non-null    str
 1   against_bug        801 non-null    float64
 2   against_dark       801 non-null    float64
 3   against_dragon     801 non-null    float64
 4   against_electric   801 non-null    float64
 5   against_fairy      801 non-null    float64
 6   against_fight      801 non-null    float64
 7   against_fire       801 non-null    float64
 8   against_flying     801 non-null    float64
 9   against_ghost      801 non-null    float64
 10  against_grass      801 non-null    float64
 11  against_ground     801 non-null    float64
 12  against_ice        801 non-null    float64
```

```
 13  against_normal     801 non-null    float64
 14  against_poison     801 non-null    float64
 15  against_psychic    801 non-null    float64
 16  against_rock       801 non-null    float64
 17  against_steel      801 non-null    float64
 18  against_water      801 non-null    float64
 19  attack             801 non-null    int64
 20  base_egg_steps     801 non-null    int64
 21  base_happiness     801 non-null    int64
 22  base_total         801 non-null    int64
 23  capture_rate       801 non-null    str
 24  classfication      801 non-null    str
 25  defense            801 non-null    int64
 26  experience_growth  801 non-null    int64
 27  height_m           781 non-null    float64
 28  hp                 801 non-null    int64
 29  japanese_name      801 non-null    str
 30  name               801 non-null    str
 31  percentage_male    703 non-null    float64
 32  pokedex_number     801 non-null    int64
 33  sp_attack          801 non-null    int64
 34  sp_defense         801 non-null    int64
 35  speed              801 non-null    int64
 36  type1              801 non-null    str
 37  type2              417 non-null    str
 38  weight_kg          781 non-null    float64
 39  generation         801 non-null    int64
 40  is_legendary       801 non-null    int64
dtypes: float64(21), int64(13), str(7)
memory usage: 256.7 KB
```

```
None
```

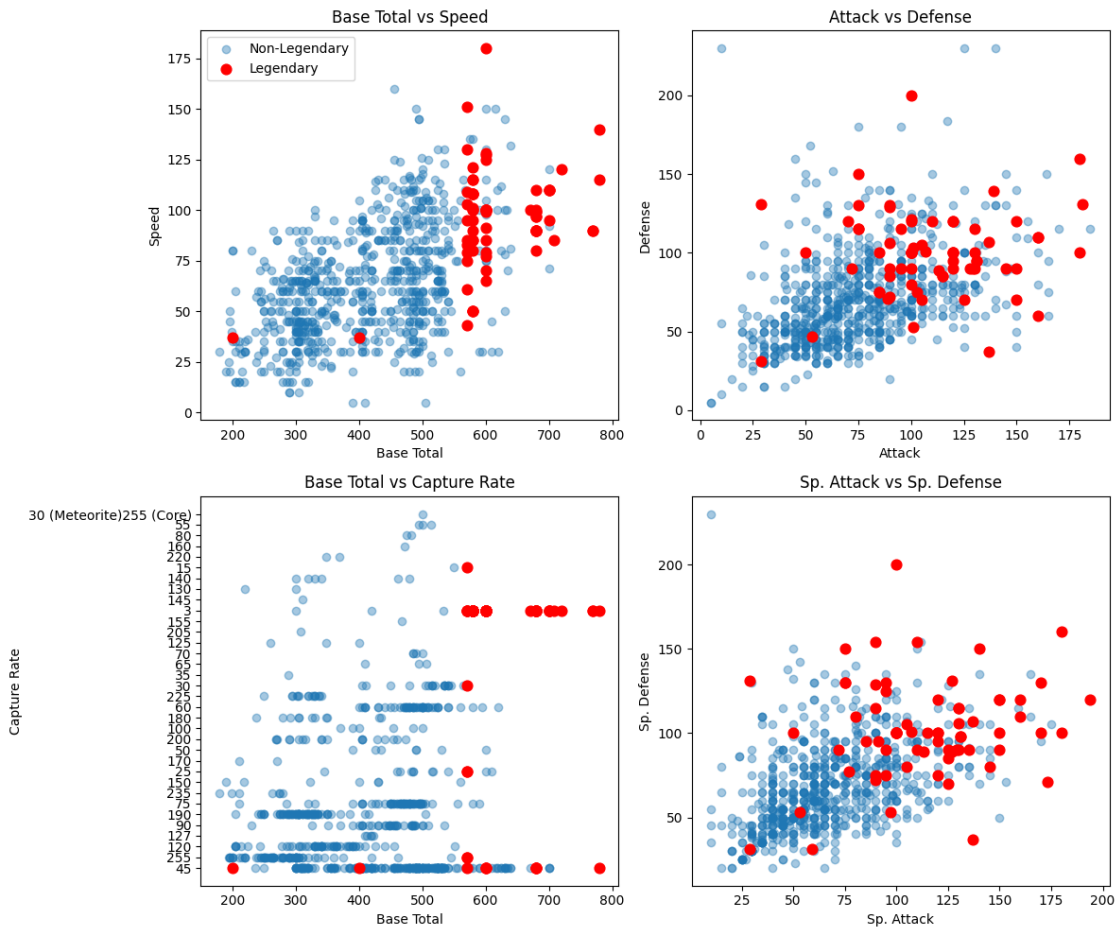
```
abilities             0.000000
against_bug           0.000000
against_dark          0.000000
against_dragon        0.000000
against_electric      0.000000
against_fairy         0.000000
against_fight         0.000000
against_fire          0.000000
against_flying        0.000000
against_ghost         0.000000
against_grass         0.000000
against_ground        0.000000
against_ice           0.000000
against_normal        0.000000
against_poison        0.000000
against_psychic       0.000000
```

```
against_rock          0.000000
against_steel         0.000000
against_water         0.000000
attack                0.000000
base_egg_steps        0.000000
base_happiness        0.000000
base_total            0.000000
capture_rate          0.000000
classfication         0.000000
defense               0.000000
experience_growth     0.000000
height_m              2.496879
hp                    0.000000
japanese_name         0.000000
name                  0.000000
percentage_male      12.234707
pokedex_number        0.000000
sp_attack             0.000000
sp_defense            0.000000
speed                 0.000000
type1                 0.000000
type2                47.940075
weight_kg             2.496879
generation            0.000000
is_legendary          0.000000
dtype: float64
```

[25]:

```
# Split data
legendary = df[df['is_legendary'] == 1]
non_legendary = df[df['is_legendary'] == 0]

fig, axes = plt.subplots(2, 2, figsize=(12, 10))
axes = axes.ravel()

# 1. Base Total vs Speed
axes[0].scatter(non_legendary['base_total'], non_legendary['speed'], alpha=0.4,␣
↪label='Non-Legendary')
axes[0].scatter(legendary['base_total'], legendary['speed'], color='red', s=60,␣
↪label='Legendary')
axes[0].set_title('Base Total vs Speed')
axes[0].set_xlabel('Base Total')
axes[0].set_ylabel('Speed')
axes[0].legend()

# 2. Attack vs Defense
axes[1].scatter(non_legendary['attack'], non_legendary['defense'], alpha=0.4)
axes[1].scatter(legendary['attack'], legendary['defense'], color='red', s=60)
```

```
axes[1].set_title('Attack vs Defense')
axes[1].set_xlabel('Attack')
axes[1].set_ylabel('Defense')

# 3. Base Total vs Capture Rate
axes[2].scatter(non_legendary['base_total'], non_legendary['capture_rate'],␣
 ↪alpha=0.4)
axes[2].scatter(legendary['base_total'], legendary['capture_rate'],␣
 ↪color='red', s=60)
axes[2].set_title('Base Total vs Capture Rate')
axes[2].set_xlabel('Base Total')
axes[2].set_ylabel('Capture Rate')

# 4. Sp. Attack vs Sp. Defense
axes[3].scatter(non_legendary['sp_attack'], non_legendary['sp_defense'],␣
 ↪alpha=0.4)
axes[3].scatter(legendary['sp_attack'], legendary['sp_defense'], color='red',␣
 ↪s=60)
axes[3].set_title('Sp. Attack vs Sp. Defense')
axes[3].set_xlabel('Sp. Attack')
axes[3].set_ylabel('Sp. Defense')

plt.tight_layout()
plt.show()
```



[26]:
```
# Cleaning

# Drop useless columns
df = df.drop(columns=[
    "abilities",
    "japanese_name",
    "name",
    "classfication"
])

# Fix data type (capture_rate is string instead of int)
df["capture_rate"] = pd.to_numeric(df["capture_rate"], errors="coerce")

# Fill type2 missing data with None, fill percentage_male, height, weight with␣
↪median
df["type2"] = df["type2"].fillna("None")
df["percentage_male"] = df["percentage_male"].fillna(df["percentage_male"].
↪median())
```

```
df["height_m"] = df["height_m"].fillna(df["height_m"].median())
df["weight_kg"] = df["weight_kg"].fillna(df["weight_kg"].median())
df["capture_rate"] = df["capture_rate"].fillna(df["capture_rate"].median())

# Encode categorical variables (type1 and type2)
df = pd.get_dummies(df, columns=["type1", "type2"], drop_first=True)
```

[27]:
```
# Define features and target variable
# Target: Is the pokemon legendary or not?
X = df.drop(columns=["is_legendary"])
y = df["is_legendary"]
```

[28]:
```
# Check no missing values left and shape
# We have 801 pokemons and 69 features and no missing values left

print("Any NaNs left:", np.isnan(X).sum().sum())
print("Shape of X:", X.shape)
```

```
Any NaNs left: 0
Shape of X: (801, 69)
```

[29]:
```
# Scale numeric features to mean=0, std=1; convert boolean features to 0/1
numeric_cols = X.select_dtypes(include=['float64', 'int64']).columns
bool_cols = X.select_dtypes(include=['bool']).columns

X[bool_cols] = X[bool_cols].astype(int)

scaler = StandardScaler()
X[numeric_cols] = scaler.fit_transform(X[numeric_cols])

print(f"Scaled {len(numeric_cols)} numeric columns, converted {len(bool_cols)}␣
↪boolean columns")
print("Shape of X:", X.shape)
```

```
Scaled 34 numeric columns, converted 35 boolean columns
Shape of X: (801, 69)
```

[30]:
```
# Train/test split (70/30, stratified by target class)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=11, stratify=y
)
print(f"Train: {X_train.shape}, Test: {X_test.shape}")
```

```
Train: (560, 69), Test: (241, 69)
```

[31]:
```
# Johnson-Lindenstrauss bound
# This is only a theoretical distance-preservation bound/ worst-case␣
↪theoretical guarantee, not a classification optimum
```

```
# If you want to preserve all pairwise distances between points with very small␣
↪error ( =0.1), then you may need a high dimension (5730 dimensions for 801␣
↪samples)
jl_bound = johnson_lindenstrauss_min_dim(n_samples=X.shape[0], eps=0.1)
print(f"Johnson-Lindenstrauss lower bound for eps=0.1: k >= {jl_bound} given␣
↪n={X.shape[0]} samples")
```

```
Johnson-Lindenstrauss lower bound for eps=0.1: k >= 5730 given n=801 samples
```

[32]:
```
# Class distribution in target variable
# Majority-class baseline accuracy (~91%) for comparison
# unbalanced classes (801 pokemons, 70 legendary → 731 non-legendary)
print(y.value_counts(normalize=True))
```

```
is_legendary
0    0.912609
1    0.087391
Name: proportion, dtype: float64
```

1.3.1 Baseline and Random Projection Experiment

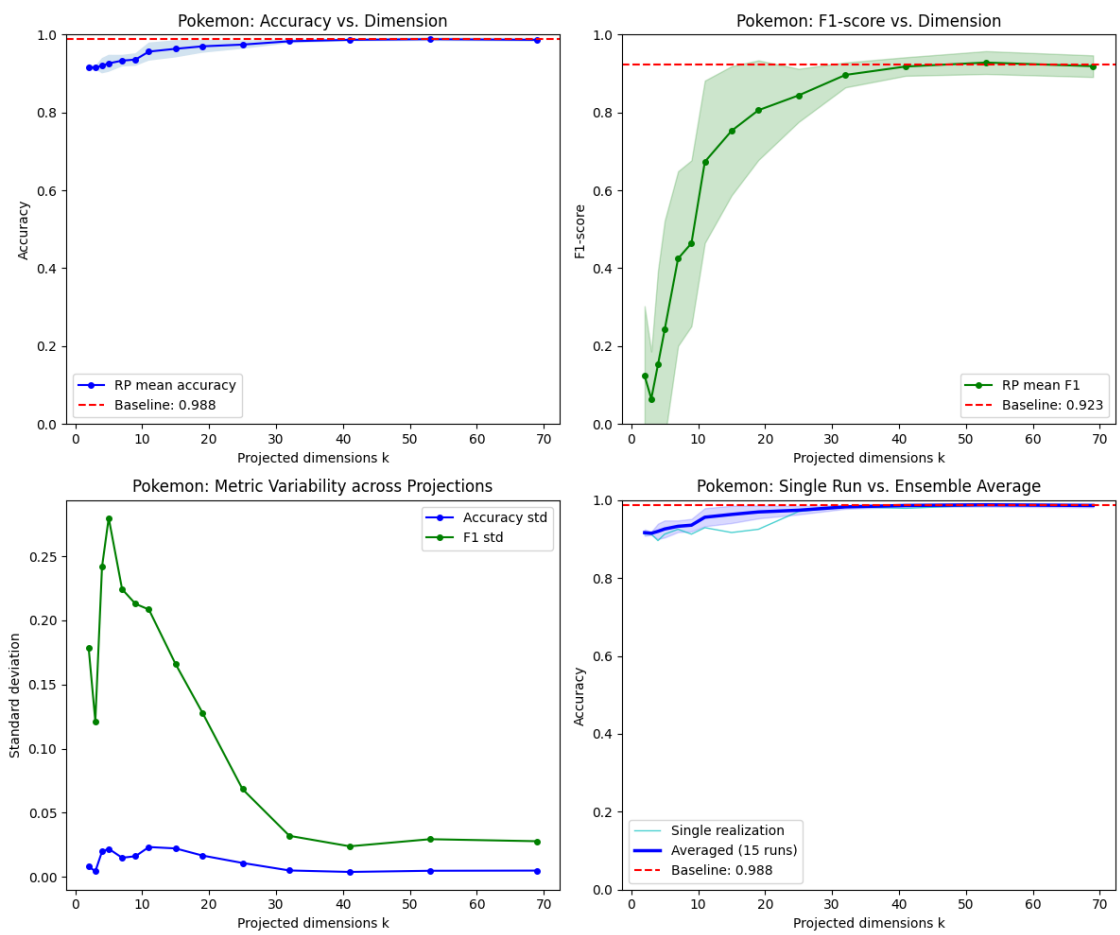
[33]:
```
# Baseline model on full dimensionality
baseline_model = LinearSVC(max_iter=1000, random_state=42)
baseline_model.fit(X_train, y_train)
baseline_pred = baseline_model.predict(X_test)

baseline_acc = accuracy_score(y_test, baseline_pred)
baseline_f1 = f1_score(y_test, baseline_pred, zero_division=0)
print(f"Baseline accuracy (d={X.shape[1]}): {baseline_acc:.4f}")
print(f"Baseline F1-score: {baseline_f1:.4f}")

# Random projection experiment (log-spaced dimensions)
realizations = 15
dims = make_dims(X.shape[1], n_points=15)
print(f"\nTested dimensions ({len(dims)}): {dims}")
print(f"Running {realizations} random projections per dimension...\n")

poke_results = run_rp_experiment(
    X_train, X_test, y_train, y_test, dims,
    realizations=realizations, max_iter=1000, f1_avg='binary'
)
```

```
Baseline accuracy (d=69): 0.9876
Baseline F1-score: 0.9231

Tested dimensions (14): [ 2  3  4  5  7  9 11 15 19 25 32 41 53 69]
Running 15 random projections per dimension…

  k=   2:  acc = 0.9167 +/- 0.0082  |  F1 = 0.1237 +/- 0.1789
```

```
k=   3:  acc = 0.9151 +/- 0.0042  |  F1 = 0.0648 +/- 0.1214
k=   4:  acc = 0.9201 +/- 0.0197  |  F1 = 0.1526 +/- 0.2416
k=   5:  acc = 0.9261 +/- 0.0219  |  F1 = 0.2422 +/- 0.2799
k=   7:  acc = 0.9331 +/- 0.0149  |  F1 = 0.4247 +/- 0.2243
k=   9:  acc = 0.9358 +/- 0.0161  |  F1 = 0.4641 +/- 0.2129
k=  11:  acc = 0.9563 +/- 0.0233  |  F1 = 0.6733 +/- 0.2085
k=  15:  acc = 0.9635 +/- 0.0222  |  F1 = 0.7529 +/- 0.1658
k=  19:  acc = 0.9698 +/- 0.0166  |  F1 = 0.8058 +/- 0.1280
k=  25:  acc = 0.9743 +/- 0.0108  |  F1 = 0.8438 +/- 0.0684
k=  32:  acc = 0.9828 +/- 0.0050  |  F1 = 0.8964 +/- 0.0320
k=  41:  acc = 0.9864 +/- 0.0039  |  F1 = 0.9180 +/- 0.0238
k=  53:  acc = 0.9881 +/- 0.0048  |  F1 = 0.9281 +/- 0.0294
k=  69:  acc = 0.9864 +/- 0.0049  |  F1 = 0.9188 +/- 0.0278
```

1.3.2 Results

[34]:
```
plot_rp_results(dims, poke_results, baseline_acc, baseline_f1,
                'Pokemon', realizations)
```



1.4 2. Fashion Dataset

Dataset: - **Fashion-MNIST** (Zalando): 60,000 grayscale images (28×28) of clothing items, 10 classes

The dataset lives in 784-dimensional Euclidean space, making it well-suited for random projection analysis.

1.4.1 Load and Explore

We load the Fashion-MNIST dataset from CSV. Each row is one image: the first column is the class label, the remaining 784 columns are pixel values (28x28 grayscale). We subsample to keep computation time reasonable.

[35]:
```
# --- Fashion-MNIST ---
fashion_df = pd.read_csv("fashion-mnist_train.csv")
# Subsample to 5,000 for faster computation
fashion_df = fashion_df.sample(n=5000, random_state=42)
fashion_data = fashion_df.iloc[:, 1:].values   # 784 pixel columns
fashion_target = fashion_df.iloc[:, 0].values   # label column

fashion_labels = {0: "T-shirt/top", 1: "Trouser", 2: "Pullover", 3: "Dress",
                  4: "Coat", 5: "Sandal", 6: "Shirt", 7: "Sneaker",
                  8: "Bag", 9: "Ankle boot"}

print(f"Fashion-MNIST: {fashion_data.shape[0]} samples, {fashion_data.shape[1]}␣
↪features, "
      f"{len(np.unique(fashion_target))} classes")
```

```
Fashion-MNIST: 5000 samples, 784 features, 10 classes
```

[36]:
```
# Visualize sample images from the dataset
fig, axes = plt.subplots(1, 10, figsize=(15, 2))

# Fashion-MNIST samples (one per class)
for i in range(10):
    idx = np.where(fashion_target == i)[0][0]
    axes[i].imshow(fashion_data[idx].reshape(28, 28), cmap=plt.cm.gray_r,␣
↪interpolation='nearest')
    axes[i].set_title(fashion_labels[i], fontsize=7)
    axes[i].axis('off')

plt.suptitle("Sample Images from Fashion-MNIST", fontsize=13)
plt.tight_layout()
plt.show()
```

1.4.2 Johnson-Lindenstrauss Bound

The JL lemma states that n points in high-dimensional space can be projected into k dimensions while approximately preserving pairwise distances, provided:

$$k \geq \frac{4\ln(n)}{\varepsilon^2/2 - \varepsilon^3/3}$$

where ε controls the distortion tolerance. This bound is very conservative; in practice, good classification accuracy can be achieved with far fewer dimensions.

[37]:
```
eps = 0.1
n_fashion = fashion_data.shape[0]

jl_fashion = johnson_lindenstrauss_min_dim(n_fashion, eps=eps)

print(f"JL minimum dimensions (eps={eps}):")
print(f"  Fashion-MNIST (n={n_fashion}): k >= {jl_fashion}")
print(f"\nOriginal dimensionality: d = 784")
print(f"The JL bound exceeds d=784, so the lemma does not guarantee␣
↪preservation at eps=0.1.")
print(f"However, JL is a worst-case bound. Classification may still work well␣
↪at much lower k.")
```

```
JL minimum dimensions (eps=0.1):
  Fashion-MNIST (n=5000): k >= 7300

Original dimensionality: d = 784
The JL bound exceeds d=784, so the lemma does not guarantee preservation at
eps=0.1.
However, JL is a worst-case bound. Classification may still work well at much
lower k.
```

1.4.3 Train/Test Split and Baseline

We split 70/30, then train a LinearSVC on the full 784-dimensional data to establish a baseline accuracy and F1-score.

[38]:
```
# Train/test split (70/30)
fash_trainX, fash_testX, fash_trainY, fash_testY = train_test_split(
    fashion_data, fashion_target, test_size=0.3, random_state=42
```

```
)

# Baseline model on full dimensionality
baseline_model = LinearSVC(max_iter=2000, random_state=42)
baseline_model.fit(fash_trainX, fash_trainY)
baseline_pred = baseline_model.predict(fash_testX)

baseline_fashion_acc = accuracy_score(fash_testY, baseline_pred)
baseline_fashion_f1 = f1_score(fash_testY, baseline_pred, average='macro',␣
↪zero_division=0)
print(f"Baseline accuracy (d={fashion_data.shape[1]}): {baseline_fashion_acc:.
↪4f}")
print(f"Baseline macro F1-score: {baseline_fashion_f1:.4f}")
```

```
Baseline accuracy (d=784): 0.7320
Baseline macro F1-score: 0.7410
```

1.4.4 Random Projection Experiment

For each target dimensionality k we generate a sparse random projection matrix, project both train and test data, train a LinearSVC, and evaluate accuracy and F1-score. We repeat this over multiple realizations and average the results.

Dimensions are log-spaced so that we test more projections at low k (where accuracy changes rapidly) and fewer at high k (where it plateaus), reducing total computation time.

[39]:
```
# Random projection experiment (log-spaced dimensions)
realizations = 15
dims_fashion = make_dims(fashion_data.shape[1], n_points=20)
print(f"Tested dimensions ({len(dims_fashion)}): {dims_fashion}")
print(f"Running {realizations} random projections per dimension...\n")

fashion_results = run_rp_experiment(
    fash_trainX, fash_testX, fash_trainY, fash_testY, dims_fashion,
    realizations=realizations, max_iter=2000, f1_avg='macro'
)
```

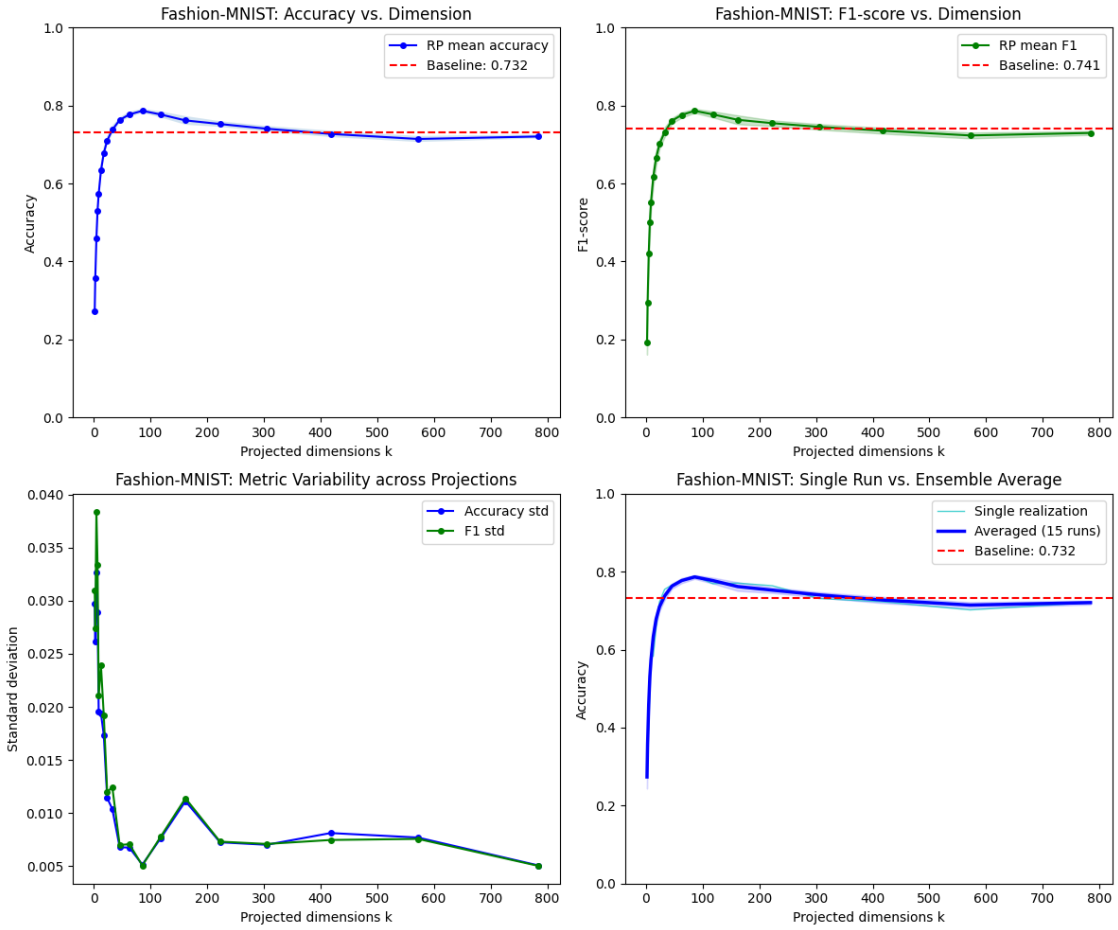
```
Tested dimensions (19): [  2   3   5   7   9  13  18  24  33  46  63  86 118 162
223 305 418 572
 784]
Running 15 random projections per dimension…

  k=   2:  acc = 0.2735 +/- 0.0298  |  F1 = 0.1920 +/- 0.0310
  k=   3:  acc = 0.3569 +/- 0.0262  |  F1 = 0.2939 +/- 0.0274
  k=   5:  acc = 0.4596 +/- 0.0327  |  F1 = 0.4204 +/- 0.0384
  k=   7:  acc = 0.5305 +/- 0.0289  |  F1 = 0.5008 +/- 0.0334
  k=   9:  acc = 0.5735 +/- 0.0196  |  F1 = 0.5520 +/- 0.0210
  k=  13:  acc = 0.6330 +/- 0.0194  |  F1 = 0.6166 +/- 0.0239
  k=  18:  acc = 0.6780 +/- 0.0173  |  F1 = 0.6658 +/- 0.0192
```

```
k=  24:  acc = 0.7103 +/- 0.0114  |  F1 = 0.7011 +/- 0.0120
k=  33:  acc = 0.7376 +/- 0.0104  |  F1 = 0.7318 +/- 0.0125
k=  46:  acc = 0.7631 +/- 0.0068  |  F1 = 0.7611 +/- 0.0070
k=  63:  acc = 0.7772 +/- 0.0067  |  F1 = 0.7759 +/- 0.0071
k=  86:  acc = 0.7866 +/- 0.0051  |  F1 = 0.7864 +/- 0.0050
k= 118:  acc = 0.7772 +/- 0.0076  |  F1 = 0.7774 +/- 0.0077
k= 162:  acc = 0.7619 +/- 0.0111  |  F1 = 0.7635 +/- 0.0114
k= 223:  acc = 0.7527 +/- 0.0072  |  F1 = 0.7548 +/- 0.0073
k= 305:  acc = 0.7406 +/- 0.0070  |  F1 = 0.7454 +/- 0.0071
k= 418:  acc = 0.7274 +/- 0.0081  |  F1 = 0.7355 +/- 0.0075
k= 572:  acc = 0.7143 +/- 0.0077  |  F1 = 0.7233 +/- 0.0076
k= 784:  acc = 0.7207 +/- 0.0051  |  F1 = 0.7298 +/- 0.0050
```

1.4.5 Results

The plots show the averaged accuracy and F1-score (solid lines) with shaded $\pm 1\sigma$ bands indicating variability across random projections. The red dashed line is the baseline on the full 784-dimensional data. The bottom-right panel demonstrates why ensemble averaging produces a smoother, more reliable curve than a single random projection.

[40]:
```
plot_rp_results(dims_fashion, fashion_results, baseline_fashion_acc,␣
↪baseline_fashion_f1,
                'Fashion-MNIST', realizations)
```



1.5 Take-Home Message

Key findings from this analysis:

1. **Random Projections act as a regularizer, not just a compressor.** The most surprising finding is that RP does not merely approximate the baseline—it can actually outperform it. For Fashion-MNIST, accuracy and F1 scores peak above the baseline between 50 and 300 dimensions. By compressing the 784 original features, RP averages out pixel-level noise and improves the samples-to-features ratio, preventing the classifier from overfitting to irrelevant details.

2. **Extreme dimensionality reduction is highly effective.** Even though the Johnson-Lindenstrauss bound suggests a higher theoretical requirement for dimensionality, empirical results show excellent performance at much lower dimensions. Fashion-MNIST achieves near-baseline (and often better) accuracy at roughly 100–200 dimensions. Similarly, the Pokémon dataset stabilizes at just 25–30 dimensions (out of 69).

3. **Ensembling is essential for stability.** A single random projection matrix yields a volatile, zig-zagging accuracy curve because each matrix preserves different structural aspects of the data. Averaging the results over multiple realizations mitigates this variance, producing

a smooth, monotonically increasing curve that clearly illustrates the relationship between dimensionality and performance.

4. **Image data in Euclidean space is inherently well-suited for RP.** Fashion-MNIST relies on Euclidean distance as a meaningful measure of similarity between images. Random projections excel here because they are mathematically guaranteed to approximately preserve these pairwise distances, maintaining the core geometry the classifier relies on to separate classes.

5. **Diminishing returns and the recovery of noise.** The accuracy curves demonstrate rapid improvement at low dimensions, peak in the mid-range (due to the denoising effect), and then converge back down to the baseline as the projection dimension approaches the original feature count. This proves that the most discriminative information lives in a significantly lower-dimensional subspace.

6. **Practical implication:** Random projections offer a powerful, data-independent method for dimensionality reduction. Unlike PCA, RP requires no expensive eigenvalue computations and can be applied to streaming data on the fly. More importantly, when tuned correctly, RP can simultaneously slash computational/storage costs *and* boost classification performance by filtering out high-dimensional noise.